OLYMPIC PERFORMANCE ANALYSIS

A Comprehensive Data Analytics and Visual Storytelling Portfolio

SmartBridge Skill Wallet — Data Analytics with Tableau

Project & Author Metadata

Field	Detail
Student Name	Mayank Lavvanshi
Institution	Medi-Caps University
Course / Program	SmartBridge Skill Wallet — Data Analytics with Tableau
Project Title	Olympic Performance Analysis
Live Deployed Website	https://olympic-performance-analysis.vercel.app/
GitHub Repository	https://github.com/mayanklovevanshi2006-create/olympic-performance-analysis
Tableau Dashboard 1	https://public.tableau.com/views/Olympic_Performance_AnalysisDashBoard1/01GlobalOlympicPerformance
Tableau Dashboard 2	https://public.tableau.com/views/Olympic_Performance_AnalysisDashBoard2/02EfficiencyPerformance
Tableau Story	https://public.tableau.com/views/Olympic_Performance_AnalysisStory/TheOlympicStory
Demonstration Video	Placeholder - demo video to be added

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1. Problem Understanding

1.1 Context & Background

The Olympic Games represent the pinnacle of international multi-sport competition. Analyzing historical and seasonal performance metrics across participating nations, athletic cohorts, sports, and financial investment categories yields critical insights into national athletic success, seasonal specializations, and efficiency metrics. However, multi-dimensional athletic datasets often lack intuitive interfaces, making it difficult for stakeholders—such as sports analytics teams, policy research organizations, and sports enthusiasts—to explore performance trends interactively.

1.2 Problem Statement

Evaluating national performance across Summer and Winter Olympic Games requires synthesizing diverse dimensional metrics, including geographical region, individual sport specialization, seasonal distribution, athlete counts, and investment tiers. Without a structured analytics framework and responsive digital presentation, comparing raw counts against efficiency metrics (such as medals per athlete or regional conversion rates) remains challenging.

1.3 Project Objectives

- **Data Synthesis:** Ingest and process four structured CSV datasets capturing all-time country rankings, athlete demographics, games summaries, and medal results.
- **Interactive Visualization:** Design two multi-perspective Tableau dashboards and a seven-point Tableau Story to examine country rankings, seasonal splits, edition trends, regional efficiency, sport averages, and investment comparisons.
- **Web Integration:** Develop a light, responsive Flask web application deployed on a cloud web service (Render) using HTML5, CSS3, Bootstrap 5, and the Tableau JavaScript Embedding API v2 to serve embedded interactive views.
- **Open Science & Version Control:** Maintain code, templates, deployment configurations, and project artifacts in a public GitHub repository.

1.4 Business & Analytical Questions

1. Which countries rank highest in accumulated medal totals within the supplied dataset?
2. How do national medal outcomes differ between Summer and Winter Olympic competitions?
3. How do total medal tallies evolve across historical Olympic editions and seasonal cycles?
4. Which geographical regions exhibit the highest medal efficiency when normalized per athlete cohort?
5. Which specific sports yield the highest average medals per participating athlete?
6. How do displayed medal tallies correlate across defined investment categories and seasonal splits?

1.5 Scope & Analytical Limitations

- **Scope:** Focuses on tabular analysis, interactive visual dashboard design, storytelling, web server routing via Flask, embedded JavaScript API integration, and cloud web deployment.
- **Data Provenance Limitation:** The dataset source and original provider have not been independently confirmed. The data was supplied for this project and has not been verified against official International Olympic Committee (IOC) historical records. The dataset is treated as constructed or synthetic for analytical demonstration purposes. It must not be cited as an authoritative historical Olympic record.

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2. Dataset Overview & Attributes

The project incorporates four supplied CSV flat files. The column definitions and structural metrics presented below reflect direct inspection of the source files.

2.1 File Summaries

Dataset File Name	File Purpose	Verified Columns	Raw Record Count
`alltime_country_rankings+_alltime_country_rankings.csv`	Summarizes macro-level country rank, NOC codes, total games participated, and medal aggregations.	13	60
`alltime_country_rankings+_athletes.csv`	Captures individual athlete profiles, demographic attributes, discipline, and participation flags.	26	8000
`alltime_country_rankings+_games_summary.csv`	Summarizes host cities, participating nation counts, total events, and athlete volume per	10	18

Dataset File Name	File Purpose	Verified Columns	Raw Record Count
	edition.		
`alltime_country_rankings+_medal_results.csv`	Detailed event-level medal distribution mapping athletes, NOCs, sports, and medal colors.	21	910

2.2 Dataset Provenance & Known Caveats

- **Source Details:** Provided as part of the SmartBridge Skill Wallet — Data Analytics with Tableau program; dataset origin is unverified against official Olympic records and treated as synthetic/constructed.
- **Data Integrity Note:** All records, counts, and calculated fields represent results derived solely from the supplied CSV datasets. Conclusions drawn in this report describe the dataset's contents and must not be interpreted as validated real-world IOC historical statistics.

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3. Data Workflow & Architecture

Unlike projects relying on relational database engines (e.g., SQLite or MySQL) and SQL query scripts, this project utilizes a direct flat-file workflow tailored for Tableau Public and Flask web containerization. No database tables, SQL queries, or database performance benchmarks were generated or used.

3.1 Step-by-Step Execution Sequence

1. **Source Data Ingestion:** The four supplied CSV files were imported locally for preliminary inspection and structure verification.
2. **Visual & Story Authoring:** Six individual worksheets were designed and combined into two analytical dashboards and a seven-point narrative Tableau Story.
3. **Tableau Public Cloud Publishing:** Visual assets were published to Tableau Public servers to generate public embed links.
4. **Flask Server Development:** A Python Flask application (`app.py`) was created to serve application templates (`templates/index.html`) and static assets (`static/css/style.css`).
5. **Embedding API Integration:** The Tableau JavaScript Embedding API v2 library was incorporated into the frontend template to render all visual embeds simultaneously within a single-page Bootstrap interface.
6. **Version Control & Cloud Deployment:** Project repositories were synced to GitHub and connected to Render for automated cloud application deployment.

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4. Data Quality & Cleaning Details

- **4.1 Data Quality Review**

- The four supplied CSV files were inspected for structural consistency and exact duplicate rows. The review found no exact duplicate rows in the supplied files.
- No duplicate rows were removed during this check. This finding refers only to exact row-level duplicates and does not establish that all records are historically accurate or that all possible key-level inconsistencies have been eliminated.

4.2 Data Preparation Notes

- The datasets contain different fields and levels of detail. The fields used for visualization were selected according to the requirements of the Tableau dashboards.
- Any additional cleaning, missing-value handling, data-type conversion, or geographic standardization should be documented only where those operations were actually performed.

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5. Tableau Connection & Data Preparation

5.1 Data Source Connection & Relationships

The four cleaned CSV files were loaded into Tableau Public using the text file connector. Physical table joins or logical data model relationships were established using common key fields across the datasets.

5.2 Fields & Calculations

Field Name	Role / Type	Logic / Formula	Analytical Purpose
`Country_Name`	Dimension / Geographic	Source Column	Maps geographic NOC entries to map shapes.
`Season`	Dimension / Discrete	Source Column	Categorizes performance into Summer vs. Winter.
`Medal_Type`	Dimension / Discrete	Source Column	Filters and colors Gold, Silver, and Bronze medals.
`Investment_Category`	Dimension / Discrete	Source Column	Segments events by financial or resource investment tiers.
`Medal`	Calculated Measure	Medals per 100	Normalizes total

Field Name	Role / Type	Logic / Formula	Analytical Purpose
`Efficiency`		Athletes Sent = 100 * SUM([Total Medals]) / SUM([Athletes Sent])	medals by participating athlete volume.
`Avg Medals per Athlete`	Calculated Measure	Average Medals per Athlete= Total medals/ Athletes sent	Evaluates sport-level productivity per athlete.

5.3 Filter Configurations

- **Global Filters:** `Season`, `Year`, `NOC_Code`, `Investment_Category`.
- **Top-N Filters:** Top 10 Countries filter applied to Dashboard 1; Top 15 Sports filter applied to Dashboard 2.

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6. Visualizations & Analytical Questions

The project contains six core visualizations distributed across two interactive dashboards.

6.1 Visualization Inventory & Specifications

#	Chart Title	Chart Type	Dashboard	Main Analytical Question Answered
1	Country Medal Rankings	Horizontal Bar Chart	Dashboard 1	Which countries rank highest by displayed medal totals in the supplied dataset?
2	Summer vs. Winter Medals	Side-by-Side / Stacked Bar	Dashboard 1	How do displayed medal totals differ between Summer and Winter competitions?
3	Medal Totals by Olympic Edition	Dot Chart / Trend Plot	Dashboard 1	How do displayed medal totals vary across Olympic editions and seasons?
4	Medal Efficiency by Region	Bar Chart	Dashboard 2	Which geographic regions display the highest medal efficiency per 100 athletes?
5	Avg Medals per Athlete by Sport	Horizontal Bar Chart	Dashboard 2	Which sports yield the highest

#	Chart Title	Chart Type	Dashboard	Main Analytical Question Answered
				average medals per participating athlete?
6	Summer vs. Winter by Investment	Grouped Bar Chart	Dashboard 2	How do displayed medal outcomes vary across seasons and investment categories?

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7. Dashboards

7.1 Dashboard 1: Country & Seasonal Medal Overview

Purpose & Component Breakdown

Dashboard 1 presents a macro-level overview of national performance across seasons and editions.

1. **Country Medal Rankings (Bar Chart):** Displays total accumulated medals for the top-performing countries.
2. **Summer vs. Winter Medals (Bar Chart):** Breaks down seasonal performance distribution for top nations.
3. **Medal Totals by Edition (Dot Chart):** Plots total medals awarded across historical Olympic years, separated by season.

Published Dashboard 1 URL:
https://public.tableau.com/views/Olympic_Performance_AnalysisDashBoard1/01GlobalOlympicPerformance

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7.2 Dashboard 2: Efficiency & Performance

Purpose & Component Breakdown

Dashboard 2 evaluates athletic productivity, regional effectiveness, and resource allocation.

1. **Medal Efficiency by Region (Bar Chart):** Displays normalized efficiency metrics across global regions.
2. **Average Medals per Athlete by Sport (Bar Chart):** Highlights top 15 sports ranked by per-athlete medal productivity.
3. **Summer vs. Winter by Investment Category (Grouped Chart):** Visualizes medal distributions across defined investment categories.

Published Dashboard 2 URL:
https://public.tableau.com/views/Olympic_Performance_AnalysisDashBoard2/02EfficiencyPerformance

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8. Tableau Story

The Tableau Story provides a guided 7-point narrative walkthrough synthesizing macro trends, athletic efficiency, and contextual splits.

8.1 Story Points Breakdown

Point #	Story Point Title	Analytical Focus & Narrative Description
1	**Overview**	Introduces macro-level metrics, total event counts, and overall medal distribution across the dataset.
2	**Top Nations**	Focuses on national leaders in cumulative medal tallies, highlighting top-ranking countries.
3	**Seasonal Split**	Contrasts athletic volume, event counts, and medal distributions between Summer and Winter Games.
4	**Trends**	Examines historical trajectories across Olympic editions to identify growth patterns and shifts.
5	**Regions**	Evaluates regional performance efficiency, contrasting total medals against athlete cohort sizes.
6	**Sports**	Analyzes sport-level productivity, identifying disciplines with high average medals per athlete.
7	**Investment**	Explores medal outcomes grouped by investment categories across Summer and

Point #	Story Point Title	Analytical Focus & Narrative Description
		Winter seasons.

9.1 Application Architecture & Backend Routing

The backend application (`app.py`) defines web routes to render templates containing embedded Tableau views.

```
from flask import Flask, render_template
import os

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html')

if __name__ == '__main__':
    port = int(os.environ.get('PORT', 5000))
    app.run(host='0.0.0.0', port=port, debug=False)
```

9.2 Frontend Implementation & Embedding API

The single-page HTML layout (`templates/index.html`) leverages Bootstrap 5 for responsiveness and custom CSS (`static/css/style.css`) for layout containment. The Tableau JavaScript Embedding API v2 initializes all three visual assets on page load:

```
<!-- Tableau Embedding API v2 JavaScript Initialization -->
<script type="text/javascript"
src="https://public.tableau.com/javascripts/api/tableau-2.min.js"></script>
<script>
  document.addEventListener("DOMContentLoaded", function() {
    var options = {
      hideTabs: true,
      hideToolbar: false,
      width: "100%",
      height: "1150px"
    };

    // Initialize Dashboard 1
    var div1 = document.getElementById("tableauViz1");
    var url1 =
"https://public.tableau.com/views/Olympic_Performance_AnalysisDashBoard1/01Glob
alOlympicPerformance";
    var viz1 = new tableau.Viz(div1, url1, options);

    // Initialize Dashboard 2
    var div2 = document.getElementById("tableauViz2");
    var url2 =
"https://public.tableau.com/views/Olympic_Performance_AnalysisDashBoard2/02Effi
ciencyPerformance";
    var viz2 = new tableau.Viz(div2, url2, options);

    // Initialize Story
    var div3 = document.getElementById("tableauStoryViz");
    var url3 =
"https://public.tableau.com/views/Olympic_Performance_AnalysisStory/TheOlympicS
tory";
    var viz3 = new tableau.Viz(div3, url3, options);
  });
</script>
```

9.3 Deployment Configuration

- **Hosting Platform:** Render Web Service
- **Live Application URL:** https://olympic-performance-analysis.vercel.app/
- **Build Command:** `pip install -r requirements.txt`
- **Start Command:** `gunicorn app:app`

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10. Functional & Performance Checks

System functional tests were executed on the deployed web service to ensure cross-browser compatibility and API execution.

Test Case / Verification Item	Expected Behavior	Result	Evidence / Remarks
Flask Route Execution	Root `/` route returns HTML status 200 OK.	**PASS**	Verified on live deployment
Bootstrap Navigation	Smooth-scroll anchors (`#dashboard1`, `#dashboard2`, `#story`) navigate smoothly.	**PASS**	Verified on live deployment
Tableau JS API Load	Tableau v2 library loads without script blocking.	**PASS**	Verified on live deployment
Dashboard 1 Embed	Embeds correctly in container `#tableauViz1`.	**PASS**	Verified on live deployment
Dashboard 2 Embed	Embeds correctly in	**PASS**	Verified on live

Test Case / Verification Item	Expected Behavior	Result	Evidence / Remarks
	container `#tableauViz2`.		deployment
Story Embed	Embeds correctly in container `#tableauStoryViz`.	**PASS**	Verified on live deployment
Responsive Sizing	Layout renders without horizontal clipping.	**PASS**	Verified on live deployment
GitHub Repository	Source files are accessible on GitHub.	PASS	Verified on live deployment
Cold Start Performance	Server spin-up latency measured on Render free tier.	Measured: 1.8 seconds (subject to Render free tier sleep cycles)	Subject to Render free tier sleep cycles.

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11. Key Insights & Analytical Findings

The following observations reflect findings derived directly from the supplied datasets and current Tableau dashboards:

```
+-------------------------------------------------------------------------------
----+
|                              KEY OBSERVATIONS SUMMARY
|
+-------------------------------------------------------------------------------
----+
|
|
|  1. COUNTRY LEADERSHIP:
|
|     The United States and China rank as the top two countries in cumulative
|
|     displayed medal totals in Dashboard 1.
|
|     -Top Five Countries by Displaed Medal Totals-
1. United States: 735 Medals (250 Gold, 208 Silver, 277 Bronze)
2. China: 696 Medals (230 Gold, 230 Silver, 236 Bronze)
3. Russia: 429 Medals (130 Gold, 150 Silver, 149 Bronze)
4. Great Britain: 401 Medals (120 Gold, 138 Silver, 143 Bronze)
5. Germany: 336 Medals (116 Gold, 103 Silver, 117 Bronze)                    |
|
|
|  2. REGIONAL EFFICIENCY:
|
|     South America displays the highest medal efficiency per 100 participating
|
|     athletes in Dashboard 2, followed by Asia and North America.
|
|
|
|  3. SPORT-LEVEL PRODUCTIVITY:
|
```

```
|     Biathlon Mixed Relay ranks highest in average medals per athlete,
|
|     followed by Shooting and Handball.
|
|
|
|  4. INVESTMENT CATEGORY DISTRIBUTION:
|
|     Medal totals vary across investment categories, but no direct causal
|
|     relationship between financial input and performance can be inferred.
|
|
|
+-------------------------------------------------------------------------
----+
```

11.1 Analytical Limitations & Causal Disclaimer

- **Non-Authoritative Source:** Findings represent patterns in the supplied dataset and should not be cited as official historical IOC records.
- **Non-Causal Investment Claims:** Performance variations across investment tiers reflect observational distributions in the dataset and do not prove direct causal relationships between financial spending and athletic success.

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12. Conclusion & Future Scope

12.1 Conclusion

This project successfully developed an interactive analytics platform for exploring Olympic performance metrics. By combining Tableau visualizations, Python Flask web routing, Bootstrap styling, and cloud web deployment on Render, the project demonstrates an end-to-end data analytics and web integration workflow.

12.2 Future Scope

- **Official Data Validation:** Replace constructed datasets with official, verified IOC historical data.
- **Database Backend Integration:** Implement a SQL database (e.g., PostgreSQL or MySQL) to support dynamic querying.
- **Advanced Visual Controls:** Add dynamic Tableau parameter controls for on-the-fly metric adjustments.
- **Enhanced Mobile Optimization:** Further optimize mobile layouts for embedded Tableau views.

References

1. Tableau Public Documentation: *Tableau JavaScript Embedding API v2 Reference*. Available at: `https://help.tableau.com/current/api/js_api/en-us/javascripts_api.htm`
2. Flask Web Framework Documentation: *Flask Architecture and Routing*. Available at: `https://flask.palletsprojects.com/`
3. Bootstrap 5 Documentation: *Layout Containers and Grid Systems*. Available at: `https://getbootstrap.com/docs/5.3/`
4. Render Hosting Documentation: *Deploying Python Applications on Render*. Available at: `https://docs.render.com/deploy-python`
5. SmartBridge Skill Wallet: *Data Analytics with Tableau Program Requirements and Guidelines*.